

AMLIS Unit-1 Home Assignment Questions:

1. For each of Supervised, Unsupervised, and Reinforcement Learning, provide a practical example and explain the differences in data requirements and learning approach.
2. Given a probability distribution of events, compute the conditional probability using Bayes' theorem and relate it to a spam email classification example.
3. Given a dataset where the price attribute has minimum and maximum values of 3,000 and 5,000 respectively, and is normalized to the range [0.5–2.0] using min-max normalization, write a Python program to transform the value 7,000 using this normalization technique. Determine through your implementation whether the resulting normalized value falls within or outside the specified range, and discuss any out-of-bound error encountered
4. suppose that the data for analysis includes the attribute age . The age values for the data tuples are (increasing order) 13,15,16,16,19,20,20,21,22, 22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46,52,70.
 - i. Use min-max normalization to transformation the value 35 for age with range [0.0,1.0]
 - ii. Use z-score normalization to transformation the value 35 for age, where the standard deviation of age is 12.94 years.
 - iii. Use normalization by decimal scaling to transform the value 35 for age.
5. Using the given dataset of hours spent driving (x): (10,9,2,15,10,16,11,16) and risk score (y): (95,80,10,50,45,98,38,93), apply Simple Linear Regression to find the equation of the best-fit line that predicts the risk score from driving hours value 9.
6. Implement a linear regression model to predict house prices from synthetic data: size, location, rooms, amenities, price; interpret the slope and intercept values.
7. Given a binary dataset of people's gender (Male / Female) as male or female from their height, then the first class could be male, implement logistic regression and compute the predicted probability for a given input.
8. Explain the working principle of the Support Vector Machine (SVM) algorithm, describing how it determines the optimal separating hyperplane. Include the concepts of margin, support vectors, and the role of kernel functions with a suitable example.
9. A COVID-19 test is conducted on 100 people:
 - Actual cases: 30 positive, 70 negative
 - Test results: TP = 25, FN = 5, TN = 60, FP = 10 ,Calculate the Accuracy, Precision, Recall, and F1-score for this test.
10. A healthcare company is designing an AI system: Predict patient recovery time, Group patients with similar symptoms, Train a surgical robot. Identify the most suitable ML type for each task and justify.

11. Using the scikit-learn framework, compute the vector addition and dot product for the given vectors $a = [4, 3]$ and $b = [2, 5]$, and interpret the results.

12. Using the scikit-learn framework, for matrices: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ — (a) Perform $A \times B$. (b) transpose of matrix and display the results.

13. Using the scikit-learn framework, implement a Python program to compute and display the gradient vector of the function $f(x,y) = x^2 + 2xy + 3y^2$ at a given point (x,y) . Demonstrate the result using at least one non-trivial input point.

14. Using a real or sample dataset containing noisy data, apply binning and clustering techniques to preprocess and improve data quality. Implement these methods using an appropriate data science framework, and display how each technique helps to reduce the impact of noise in your results.

15. Given a dataset containing a 'Salary' attribute with minimum value ₹20,000 and maximum value ₹80,000, and knowing that the mean is ₹45,000 and the standard deviation is ₹10,000, calculate the z-score normalized value for any given salary. Using your program, normalize a sample value (e.g., ₹60,000). Based on your results, explain practical scenarios where z-score normalization is preferred over other techniques.

16. Demonstrate different types of Exploratory Data Analysis (EDA) techniques: such as univariate analysis, bivariate analysis, and multivariate analysis.

17. Explain the theory behind logistic regression as you implement the model, interpret the coefficients, and evaluate the model's performance on the dataset.

18. Given Dataset as Hours Studied: 2, 4, 6, 8; Exam Passed: 0, 0, 1, 1 — Use linear regression to estimate coefficients & Predict for 5 hours studied.

19. A logistic regression model gives $P(y=1|x) = 0.72$. (a) Classify with threshold = 0.7. (b) Explain recall effect if threshold lowered to 0.5.

20. A quality control test is conducted on 100 manufactured products: Actual defective products: 20 defective, 80 non-defective

Test results: True Positives (TP) = 15, False Negatives (FN) = 5, True Negatives (TN) = 70, False Positives (FP) = 10.

Calculate the Accuracy, Precision, Recall, and F1-score for this quality control test.

21. A new diagnostic tool for detecting a rare genetic disorder was tested on a group of 200 individuals. Out of these, 50 actually had the disorder (positive class), and 150 did not (negative class). The tool produced the following results: True Positives (TP): 40, False Negatives (FN): 10, True Negatives (TN): 135, False Positives (FP): 15. Apply your understanding of evaluation metrics to compute the following for this diagnostic tool: Accuracy, Precision, Recall (Sensitivity), F1-score.